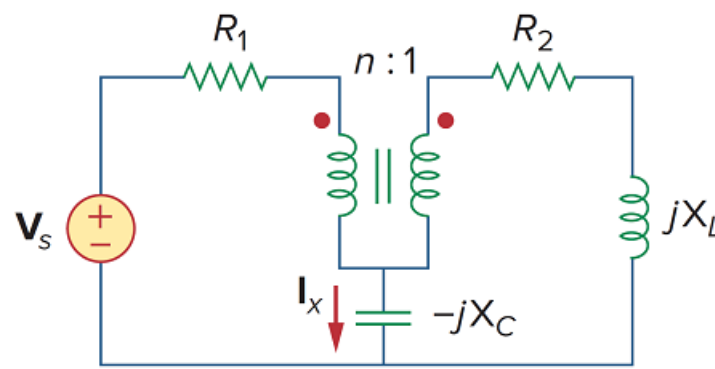


Problem

Using Fig. 13.113, design a problem to help other students better understand how ideal transformers work.

Figure 13.113



Step-by-step solution

Step 1 of 6

Refer to Figure 13.113 in the textbook.

Draw the modified diagram as shown in Figure 1.

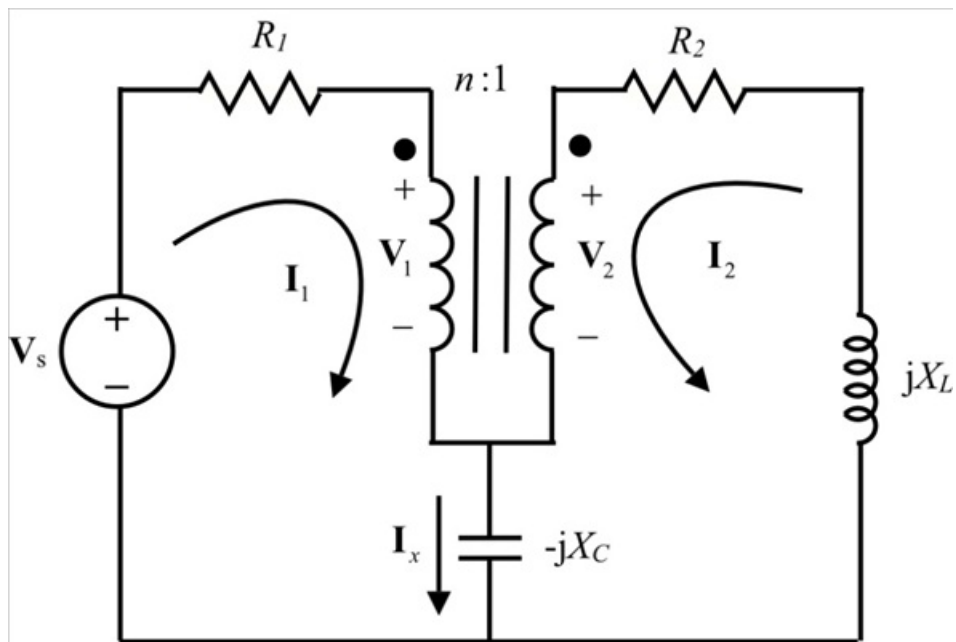


Figure 1

Step 2 of 6

Apply mesh analysis in loop 1.

$$\mathbf{V}_s = R_1 \mathbf{I}_1 - jX_C (\mathbf{I}_1 + \mathbf{I}_2) + \mathbf{V}_1$$

$$\mathbf{V}_s = (R_1 - jX_C) \mathbf{I}_1 - jX_C \mathbf{I}_2 + \mathbf{V}_1 \dots\dots (1)$$

Apply mesh analysis in loop 2.

$$0 = R_2 \mathbf{I}_2 + jX_L \mathbf{I}_2 - jX_C \mathbf{I}_1 + \mathbf{V}_2$$

$$(R_2 + jX_L) \mathbf{I}_2 - jX_C \mathbf{I}_1 + \mathbf{V}_2 = 0 \dots\dots (2)$$

Step 3 of 6

From Figure 1,

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{N_1}{N_2}$$

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{n}{1}$$

$$\mathbf{V}_1 = n\mathbf{V}_2 \dots\dots (3)$$

From Figure 1,

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_2}{N_1}$$

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{1}{n}$$

$$\mathbf{I}_2 = n\mathbf{I}_1 \dots\dots (4)$$

Step 4 of 6

Replace equation 3 and equation 4 in equation 1 and equation 2.

$$\mathbf{V}_s = (R_1 - jX_C) \mathbf{I}_1 - jX_C \mathbf{I}_2 + \mathbf{V}_1$$

$$\mathbf{V}_s = (R_1 - jX_C) \mathbf{I}_1 - jX_C \mathbf{I}_2 + n\mathbf{V}_2 \dots\dots (5)$$

$$(R_2 + jX_L)(n\mathbf{I}_1) - jX_C \mathbf{I}_1 + \mathbf{V}_2 = 0$$

$$(R_2 n + jnX_L - X_C) \mathbf{I}_1 + \mathbf{V}_2 = 0$$

$$\mathbf{V}_2 = -(R_2 n + jnX_L - X_C) \mathbf{I}_1 \dots\dots (6)$$

Step 5 of 6

Replace equation 6 in equation 5.

$$\begin{aligned} \mathbf{V}_s &= (R_1 - jX_C) \mathbf{I}_1 - jX_C (n\mathbf{I}_1) + n(-(R_2 n + jnX_L - X_C) \mathbf{I}_1) \\ &= (R_1 - jX_C) \mathbf{I}_1 - jnX_C \mathbf{I}_1 + n(-R_2 n - jnX_L + X_C) \mathbf{I}_1 \\ &= R_1 \mathbf{I}_1 - jX_C \mathbf{I}_1 - jnX_C \mathbf{I}_1 - n^2 R_2 \mathbf{I}_1 - jn^2 X_L \mathbf{I}_1 + nX_C \mathbf{I}_1 \\ &= [(R_1 - n^2 R_2 + nX_C) - j(X_C + nX_C + n^2 X_L)] \mathbf{I}_1 \end{aligned}$$

$$\mathbf{I}_1 = \frac{\mathbf{V}_s}{[(R_1 - n^2 R_2 + nX_C) - j(X_C + nX_C + n^2 X_L)]}$$

Step 6 of 6

Replace $\mathbf{I}_1$ value in equation 4.

$$\mathbf{I}_2 = n \left(\frac{\mathbf{V}_s}{\left[\left(R_1 - n^2 R_2 + nX_C \right) - j \left(X_C + nX_C + n^2 X_L \right) \right]} \right)$$

Calculate the current $\mathbf{I}_x$.

$$\begin{aligned} \mathbf{I}_x &= \mathbf{I}_1 + \mathbf{I}_2 \\ &= \frac{\mathbf{V}_s}{\left[\left(R_1 - n^2 R_2 + nX_C \right) - j \left(X_C + nX_C + n^2 X_L \right) \right]} \\ &\quad + n \left(\frac{\mathbf{V}_s}{\left[\left(R_1 - n^2 R_2 + nX_C \right) - j \left(X_C + nX_C + n^2 X_L \right) \right]} \right) \\ &= \left[\frac{\mathbf{V}_s}{\left[\left(R_1 - n^2 R_2 + nX_C \right) - j \left(X_C + nX_C + n^2 X_L \right) \right]} \right] (1+n) \end{aligned}$$

Therefore, the value of the $\mathbf{I}_x$ is $\boxed{\left[\frac{\mathbf{V}_s}{\left[\left(R_1 - n^2 R_2 + nX_C \right) - j \left(X_C + nX_C + n^2 X_L \right) \right]} \right] (1+n)}.$